

# Eating an apple:

First bite: one whole worm in your apple

→ Feel disgusted

$\frac{1}{2}$  of the worm  $\Rightarrow$  disgust more

| <u>Fraction of worm in the apple</u> |   | <u>Level of Disgust</u> |
|--------------------------------------|---|-------------------------|
| 1                                    | → | 5                       |
| $\frac{1}{2}$                        | → | 10                      |
| $\frac{1}{3}$                        | → | 15                      |
| $\frac{1}{4}$                        | → | 20                      |
| ↓                                    |   | ↓                       |
| 0                                    |   | $\infty$                |
| No worm = 0                          | → | 0                       |

□ 1)  $\underline{AB} = \underline{BA}$  does not hold in general

ii)  $\underline{AB} = \underline{0}$  (null matrix)

→ all the entries = 0

( $ab=0$   
 $\Rightarrow a=0$   
 $b=0$ )

Still none of them can be 0

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \underline{A^2=0}$$

$\neq 0 \quad \neq 0 \quad \rightarrow \quad 0$

$$\begin{matrix} \text{A} \searrow \\ \begin{matrix} \textcircled{X} \rightarrow AX \\ \textcircled{X} \leftarrow A \rightarrow XA \end{matrix} \end{matrix}$$

□ Let  $A, B$  both  $2 \times 2$  real matrices. Show that

$$C^2 = (\underline{AB - BA})^2 \quad [C = AB - BA]$$

commutes with all  $2 \times 2$  matrices.

$$C \neq X, 2 \times 2, \quad C^2 X = X C^2$$

1st step:  $\underline{\text{Tr}(C) = 0}$   $[\text{Tr}(CAB) = \text{Tr}(CBA)]$

2nd step:  $C, 2 \times 2$  matrix,  $\underline{\text{Tr}(C) = 0}$

char eq  $\rightarrow C^2 = kI \quad C \in \mathbb{R}$

$$C^2 - (\text{Tr}(C))C + \det(C)I = 0$$

$$\Rightarrow C^2 = -\det(C)I = \underline{kI}$$

3rd step: for any  $X, 2 \times 2$

$$\begin{matrix} C^2 X = kIX = kX \\ X C^2 = kX \cdot I = kX \end{matrix} \} \text{ equal}$$

□ If  $\underline{AB = BA}$ , can we say,

$$\underline{A^m B^n = B^n A^m}$$

(Any power of A, any power of B commute?)

$C, A, B^2$   
commute.  $AB^2 = (AB)B = (BA)B$  (Associative)  
 $= B(AB)$   
 $= B(BA)$   
 $= B^2A$

1st step:

$A^k, B$  commute for all  $k \in \mathbb{N}$ .  
 $\downarrow$   
 easily by induction.

2nd step:

Fix  $m, A^m, B^n$  commute,  $\forall n \in \mathbb{N}$

Base Case:  $A^m, B$  commute

$\hookrightarrow A^m, B^n$  commute for all  $m, n \in \mathbb{N}$ .

$\square$   $A, B$  commute,

Can you say  $(AB)^n = A^n B^n$ , for all  $n \in \mathbb{N}$ ?

Yes By induction.

$\square$  Inverses:

Idempotent Matrix  $\leftarrow I$

$A^2 = A$

Let  $A$  be an idempotent matrix. Then which of the following is/are always invertible?

$(I - \frac{1}{2}A)$   
 $(I - \frac{1}{2}A)(I + A) = I$

$(A) I + A$   
 $(I + 2A)$

$(I - 2A)$   
 $(I - A)$

$A^2 = A \Rightarrow A^3 = A^2 A = A$   
 $A^4 = A^3 A = A^2 A = A$

Neumann Sum

$|c| < 1$

$(I - \frac{2}{3}A)$

$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$   
 $(1-x)^{-1} = 1 + x + x^2 + \dots$

$(I + cA)^{-1}$

$= I - cA + c^2 A^2 - c^3 A^3 + \dots$   
 $= I - cA + c^2 A - c^3 A + \dots$   
 $= I - cA(1 - c^2 + c^3 - \dots)$   
 $= I - cA(1 - c + c^2 - \dots)$   
 $= I - cA \frac{1}{1-c} = I - \frac{c}{1-c} A$

Similarly

$(I - cA)^{-1}$

$= I + cA + c^2 A^2 + \dots$   
 $= I + cA(1 + c + c^2 + \dots)$   
 $= I + \frac{c}{1-c} A \Rightarrow (I + \frac{c}{1-c} A)^{-1} = I - cA$

$\frac{c}{1-c} = 1$   
 $\Rightarrow c = 1 - c$   
 $\Rightarrow c = \frac{1}{2}$

$(I + 2A)^{-1} = (I - \frac{2}{3}A)$

$\frac{c}{1-c} = 2 \Rightarrow c = 2 - 2c$   
 $\Rightarrow 3c = 2$   
 $\Rightarrow c = \frac{2}{3}$

check

$(I + 2A)(I - \frac{2}{3}A)$   
 $= I - \frac{2}{3}A + 2A - \frac{4}{3}A^2$   
 $= I + 2A - \frac{2}{3}A - \frac{4}{3}A$   
 $= I + 2A - 2A = I$

$$(I - 2A)^2 = I - 4A + 4A^2 = I - \cancel{4A} + \cancel{4A} = I$$

$$\square A^2 = A, \text{ inverse of } I + 5A.$$

$$\begin{aligned} (I - \frac{5}{6}A)(I + 5A) \\ = I - \frac{5}{6}A + 5A - \frac{25}{6}A \\ = I + \cancel{5A} - \cancel{5A} = I. \end{aligned}$$

$$\square \text{ Nilpotent Matrix : } \boxed{A^k = 0} \quad (\exists k \in \mathbb{N})$$

$$A^2 = 0$$

$$A^n = 0, \quad \forall n \geq k$$

$$(I - A)^{-1} = I + A$$

$$\downarrow$$

$$\underline{I + A + A^2 + A^3}$$

$$\square A^3 = 0.$$

$$(I - A)^{-1} = I + A + A^2$$

$$\hookrightarrow \underline{I + A + A^2 + A^3} = 0$$

$$(I - A)(I + A + A^2)$$

$$= I - A^3 = I - 0 = I.$$

$$\square A^3 = 0. \quad \text{Inverse of } I + 5A = ?$$

$$(I - 5A + 25A^2)$$

$$(I + 5A)(I - 5A + 25A^2) = I - 125 \frac{A^3}{=0} = I.$$

$$\square \text{ Let,}$$

$$A + B + AB = 0.$$

$$(A, B, n \times n)$$

Do A, B commute?

Yes

$$\begin{cases} IPB = I \\ \text{BOP} = I \end{cases}$$

$$I + A + B + AB = I$$

$$\Rightarrow (I + A)I + (I + A)B = I \Rightarrow (I + A)(I + B) = I.$$

$$\Rightarrow (I + B)(I + A) = I$$

$$\Rightarrow I + A + B + BA = I$$

$$\Rightarrow \cancel{A+B} + BA = 0 = \cancel{A+B} + AB$$

$$\Rightarrow BA = AB \quad \square$$

$\square$  Let A, B two square matrices of the same order, and  $A^2 = A, B^2 = B$

Suppose further

$$(A+B)^2 = A+B, \text{ then}$$

$$\checkmark A) \quad \underline{A, B \text{ Commute}}$$

$$\checkmark B) \quad AB = 0 \quad \checkmark$$

$$\checkmark C) \quad BA = 0 \quad \checkmark$$

$$\boxed{\text{D) All of these.}}$$

$$AB = -BA$$

$$AB = BA$$

$$\overset{A}{A^2} + \overset{B}{AB} + \overset{B}{BA} + \overset{B}{B^2} = \overset{A+B}{A+B}$$

$$\Rightarrow AB = -BA$$

$$\Rightarrow A(AB) = -ABA$$

$$\Rightarrow A^2B = -(-BA)A$$

$$\Rightarrow A^2B = BA^2$$

$$\Rightarrow \underline{AB = BA}$$

□ (H.W.)

Let  $A, B$  square matrices with  $A$  is invertible

$$AB + BA = D$$

Then show that:

$$A^{-1}B + BA^{-1} = D \quad \checkmark$$

—————/—————.